Prakhar Shrivastava

Gen AI Engineer

Gwalior, Madhya Pradesh, India | prakharshrivastava290@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Gen AI Engineer and B.Tech Artificial Intelligence student with hands-on experience building LLM-powered agents, RAG pipelines, and production-ready ML services using Python, FastAPI, and vector databases. Experienced in designing MCP-based tool-calling frameworks and delivering measurable results, including a 94.4% reduction in HVAC energy consumption through an autonomous AI agent.

Education

Vellore Institute of Technology May 2023 – May 2027 (Expected)
Bachelor of Technology in Artificial Intelligence **CGPA: 7.84**

Technical Skills

Programming & Querying: Python, SQL, Rust, Data Manipulation, ETL Processes
Machine Learning: MLOps, Scikit-learn, Feature Engineering, Classification, Clustering, Time Series Analysis
LLM & Agentic AI: Generative AI, RAG, Prompt Engineering, Agentic AI, MCP Protocol, LLMs, LangChain
Databases: Oracle AI Vector DB, PostgreSQL, MySQL, Database Design & Optimization

Experience

AIML Intern May 2026 – July 2026
STAQO *On-Site, Noida*

- Engineered an NLP-based complaint routing engine achieving 87% classification accuracy, and built FastAPI services for an Employee Grooming Engine supporting 4 workflows.
- Developed an ML-based health-checking engine for Sleepwell banner creatives with 83% accuracy, and integrated a chatbot with a behaviour-based quiz system into the Employee LMS Portal, enhancing interactive and adaptive learning.

Projects

Athena AI – Autonomous HVAC Control Agent Link
Python, EnergyPlus, Qwen2.5 LLM, Streamlit

- Build an autonomous agent to cut HVAC energy consumption in buildings while preserving occupant thermal comfort.
- Orchestrated a closed-loop AI agent by linking live EnergyPlus simulations with Qwen2.5 LLM through a custom MCP tool-calling framework.
- Achieved a 94.4% reduction in HVAC energy consumption (250,734 kWh saved) while maintaining 100% occupant thermal comfort across 676+ autonomous control timesteps.

Docsy – Grounded Multi-PDF RAG Assistant Link
Python, LangChain, FAISS, HuggingFace, Groq, Streamlit

- Engineered a multi-PDF RAG pipeline using LangChain, HuggingFace, and FAISS with recursive chunking, stale-data detection, and page-level cross-document citations.
- Built low-latency inference with Groq, with strict anti-hallucination prompt guardrails, achieving sub-second responses and zero hallucination on out-of-scope queries.
- Developed a Streamlit app with a retrieval k-slider, real-time ingestion progress and a document catalog; validated with a 69-test PyTest suite using mocked dependencies, ensuring zero regression at no API cost.

Achievements

Vultr Cloud Innovate Hackathon

- Developed a cloud-based analytics solution leveraging AI/ML for automated data processing and insight generation; presented data-driven recommendations to judges.

ArthNeeti 4.0 Economic Analysis

- Reached the Finals and secured 7th place nationally; conducted statistical analysis and data modeling to evaluate economic policies using quantitative methods and visualization techniques.

Certifications

- Oracle AI Vector Search Professional
- Oracle Generative AI Professional